

OUTDOOR ROOMBA (RUMBLE)

TEAM ID: SDDEC25-16

Advisor's name: Dr. Diane Rover





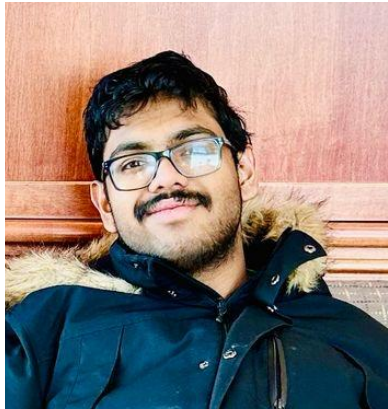
Mitchell Owen: Electrical Engineer, Tester



Brodie Bates: *Embedded Engineer, Designer*



Daniel Vergara Pinilla: *Machine Learning Engineer*



Sudipta Halder: *Embedded Engineer, Designer*



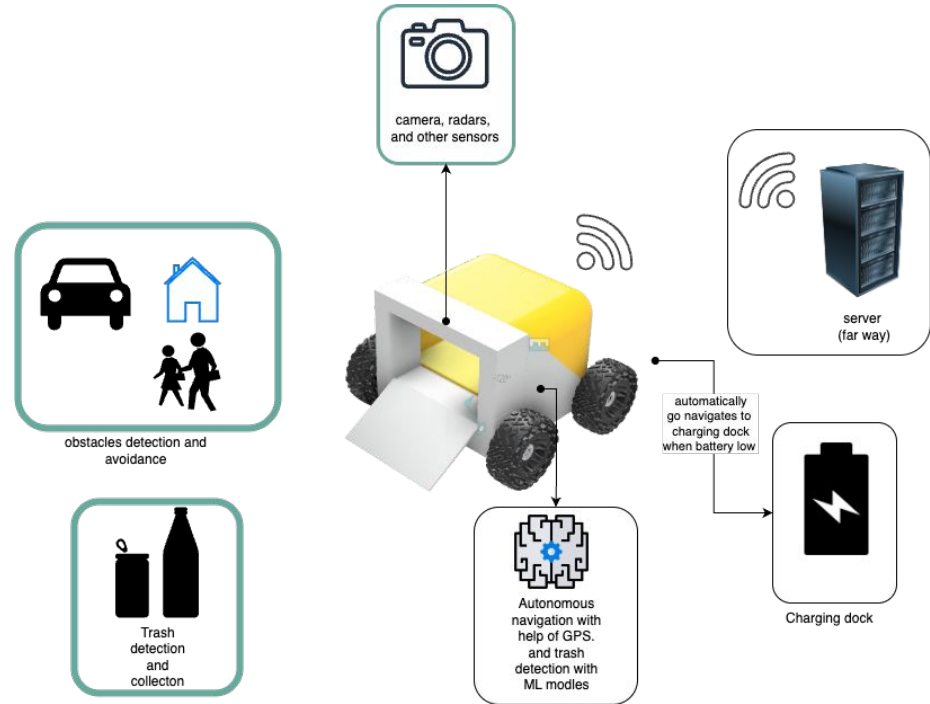
Om Shukla: *Front-end Engineer*

Rumble Robot

Roomba that picks up trash

Bumble: Outdoor Roomba

1. Autonomous
2. Object And Trash Recognition
3. Refuse Collector
4. Unified Endpoint Management



Problem and Users

- Problem:
 - 50 Billion Pieces of Litter
- Users
 - Event Managers
 - Family Gathering
 - Public Locations
 - Private Events
 - Volunteers/Nonprofits
 - Keep America Beautiful
 - Pick-Up Iowa

Functional Requirements

- Clean a Given Area
 - Navigate in Urban Areas
 - Statistic Tracking
 - Object Identification
- Accessible: Easy Maintenance
 - Low Cost
 - Replaceable Parts
 - Easily Manage Multiple bots
- Safety: Avoid Collisions
 - Navigate Around People

Non-Functional Requirements

- Clean a Given Area
 - Detect Obstacles
 - Track Trash in the Area
 - Carry 5 lbs and pick up 1 lb of Trash
 - Use A Server And User Interface To Manage Robot/Robots
- Accessible: Easy Maintenance
 - 3D Printable
 - Accessible Parts
 - Under 500\$
- Safety: Avoid Collisions
 - Stop On Command
 - Use SLAM and Sensors(Lidar, Ping)

Resource Requirements

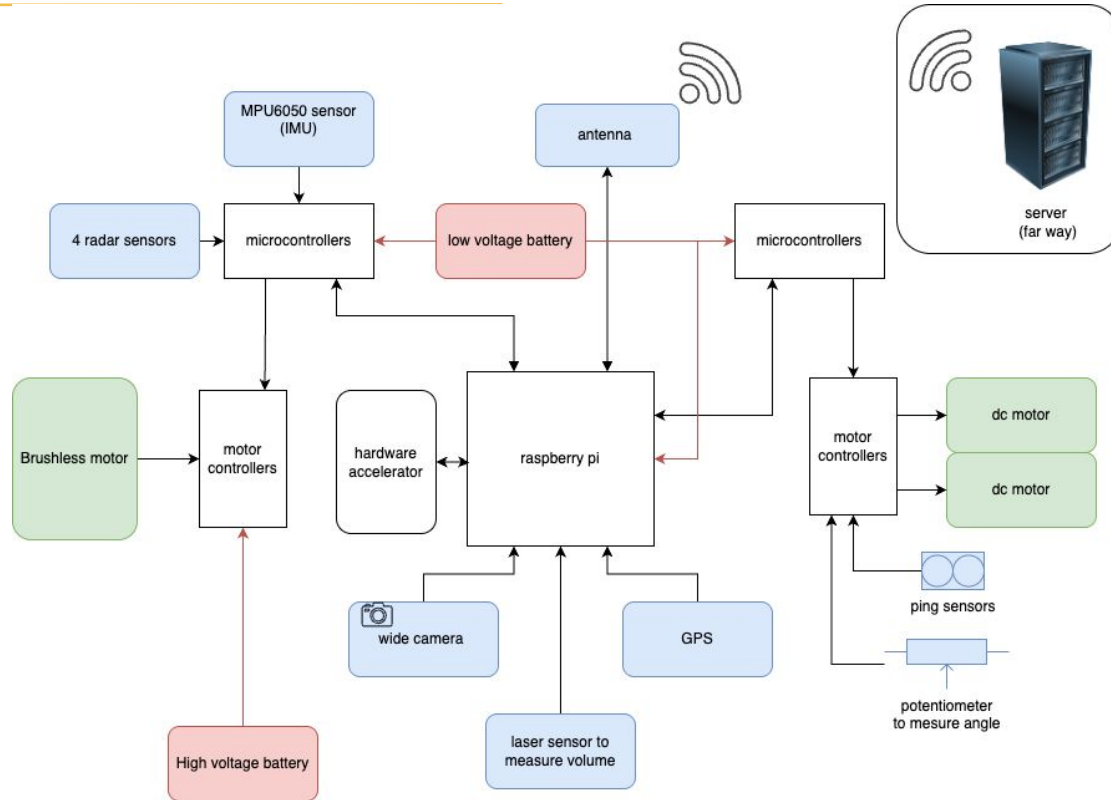
- Electronics
 - Sensors and Actuator Controller
 - Microcontrollers
 - Machine Learning Accelerator
 - Dedicated Server
- Materials
 - PLA
 - Aluminum Plate
 - Continues Crystal Acrylic
 - Wood Planks for prototyping
- Equipment
 - 3D Printer
 - Ultimaker
 - Oscilloscopes
 - Dual 500 MHz digital storage oscilloscope
 - Multimeters
 - Power Supplies
 - ABE's Waterjet

Risks and Mitigation

Task	Risk	Chance	Risk Level	How to Fix It
Designing the Robot Chassis	Weak structure	40%	Moderate	CAD simulations, stress testing
	Poor weight distribution	50%	Moderate	Prototype different configurations
	Manufacturing constraints	30%	Low	Consider a premade chassis
Building the Physical Robot	Assembly difficulties	50%	Moderate	Use modular design, precise fabrication
	Durability issues in outdoor conditions	60%	High	Use rugged materials, conduct environmental testing
	Motor and power inefficiencies	40%	Moderate	Test power requirements, use efficient motors
Designing the Embedded System (Sensors)	Sensor integration issues	60%	High	Choose compatible sensors, test connections early
	Signal interference or calibration errors	50%	Moderate	Use shielding and recalibrate regularly
	High power consumption	40%	Moderate	Optimize sensor usage, low-power components
Programming the Embedded System (Sensors)	Data processing lag	60%	High	Optimize code, reduce unnecessary computations
	Software bugs leading to erratic behavior	50%	Moderate	Test modules separately before full integration
Machine Learning for Object Detection	Insufficient training data	70%	High	Use pre-trained models, collect real-world data
	Slow processing speed for real-time detection	60%	High	Offload processing to edge/cloud computing
Server Infrastructure & Communication	Latency in cloud communication	50%	Moderate	Use edge computing for real-time processing
	UI usability issues	40%	Moderate	Conduct user testing and feedback loops

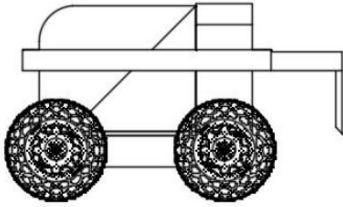
DESIGN

Rumble Systems

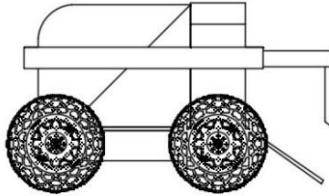


Pickup Mechanism

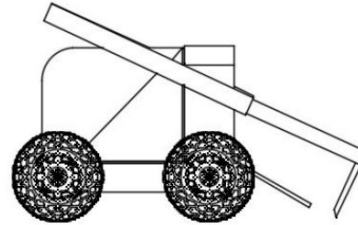
1. Start



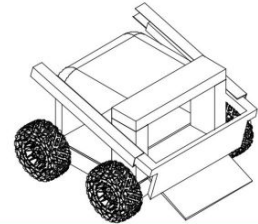
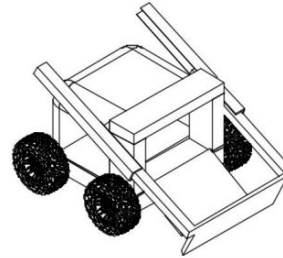
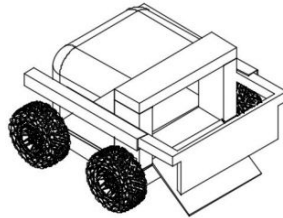
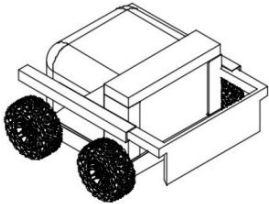
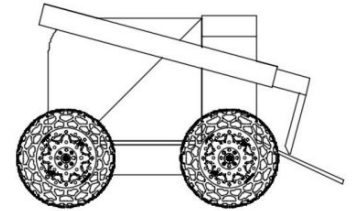
2. Shovel Drop



3. Extend



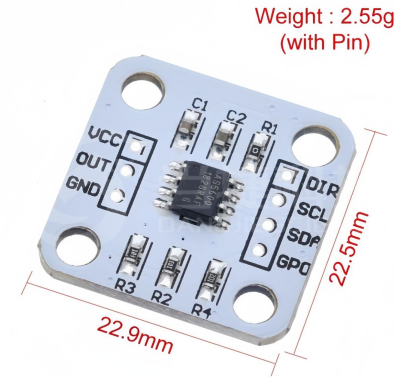
4. Pull



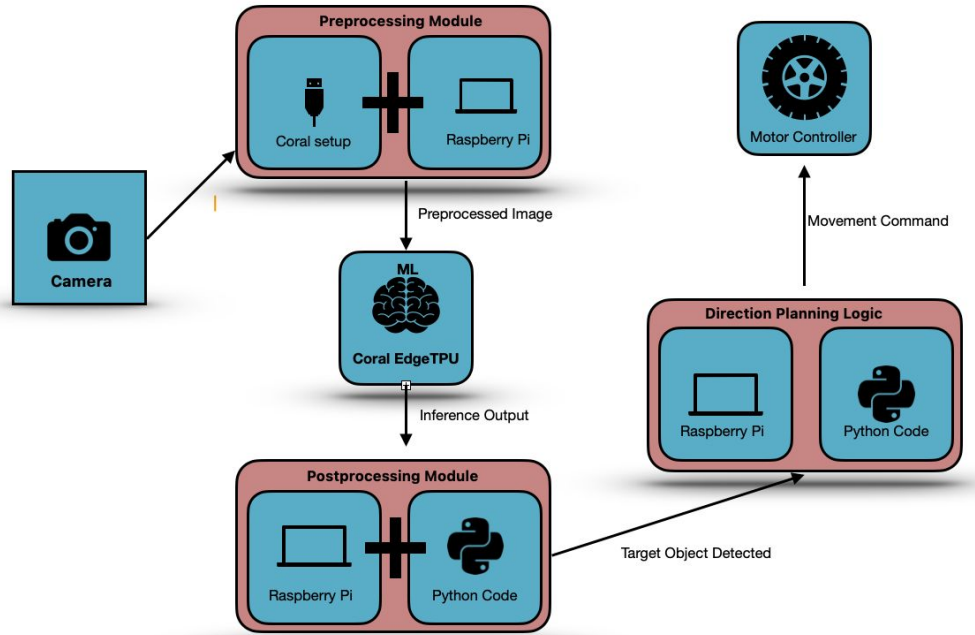
Pickup Mechanism Video

Sensors

- LD14P Lidar
- PING Ultrasonic Distance Sensor
- AS5600 Magnetic Encoder High Precision Sensor Module, 12 bit

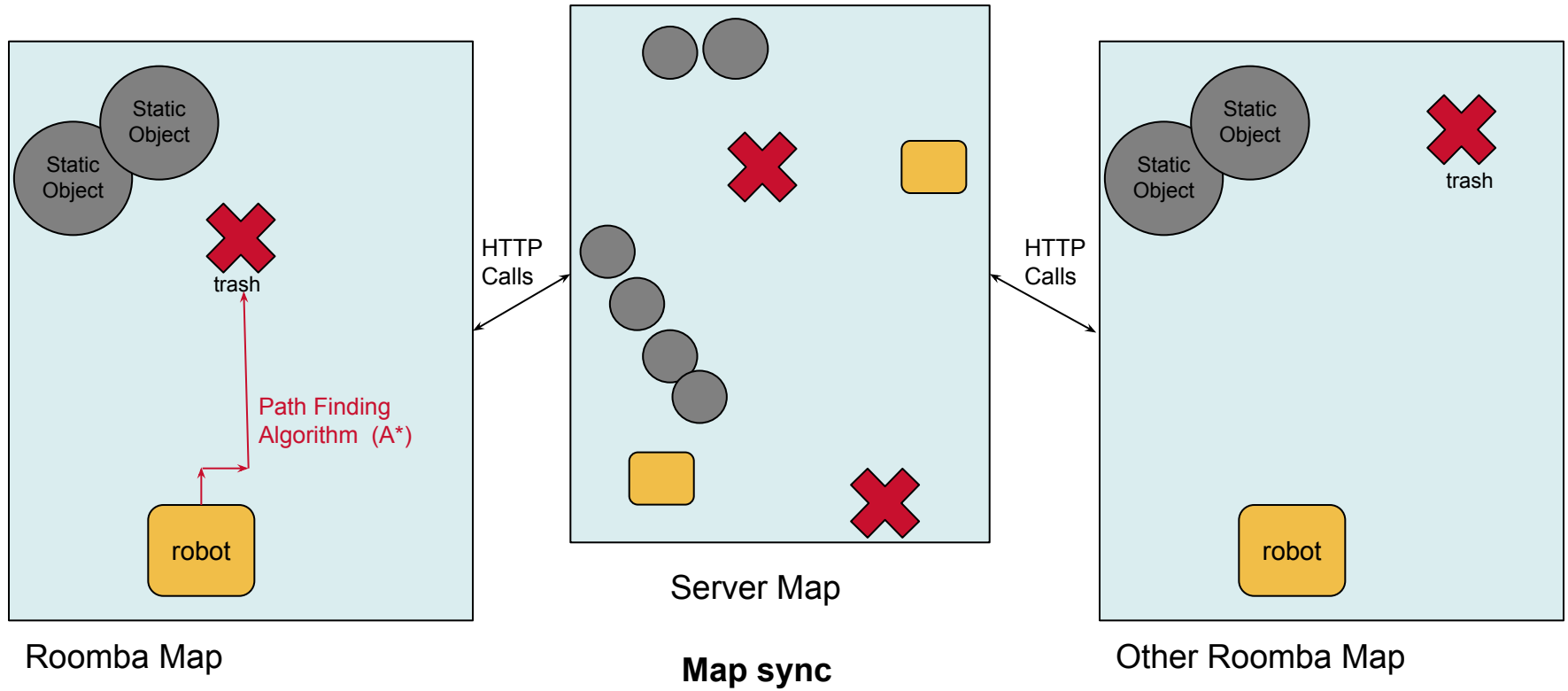


Software Design: Machine Learning

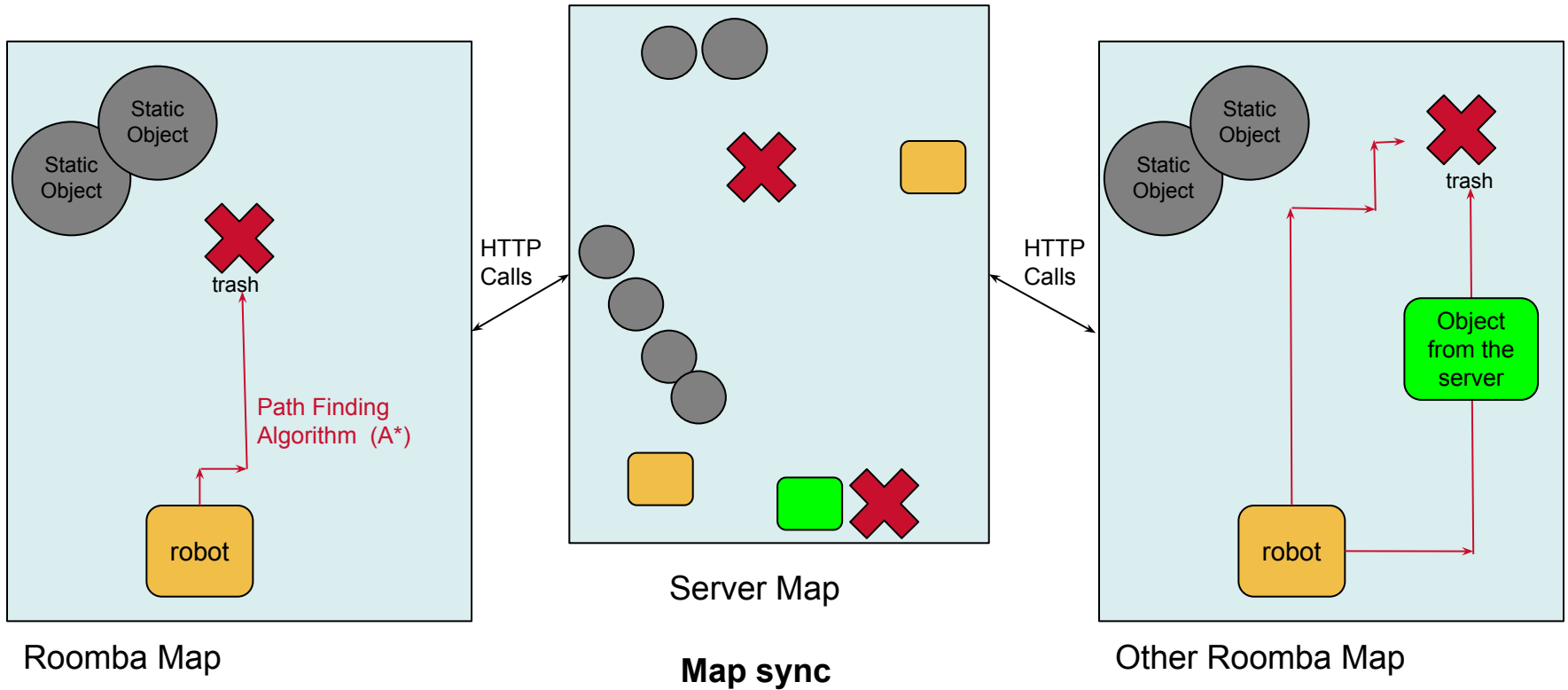


Object Detection Video

SLAM

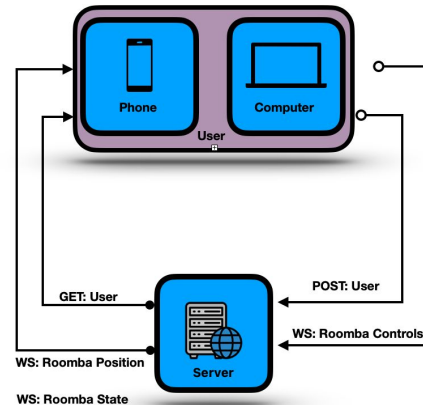
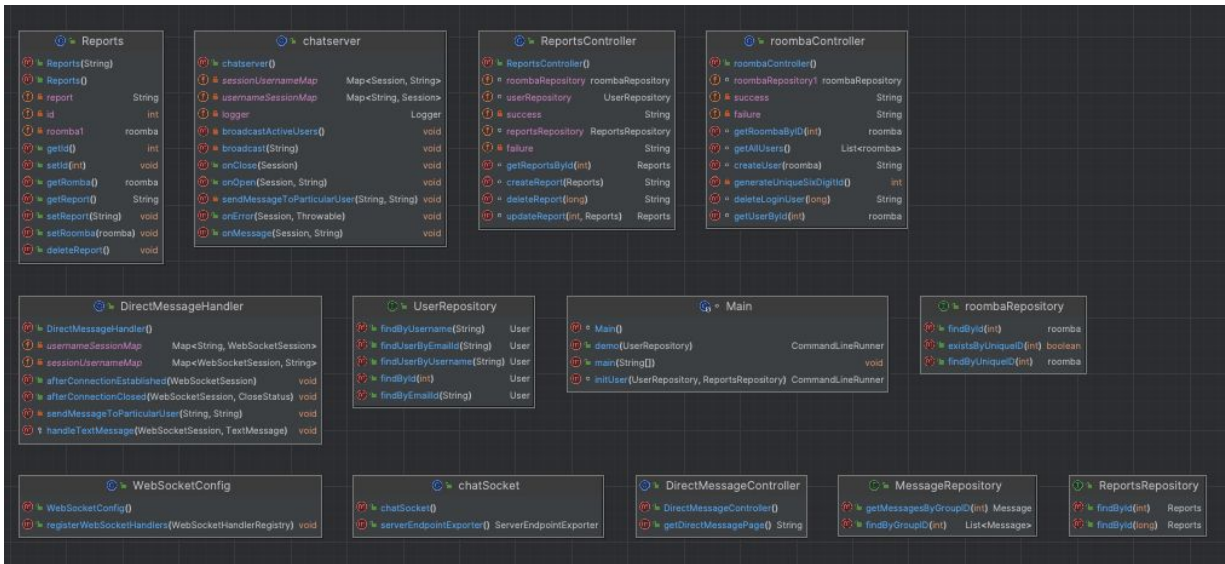


SLAM (cont)



Driving Video

Software Design: Server

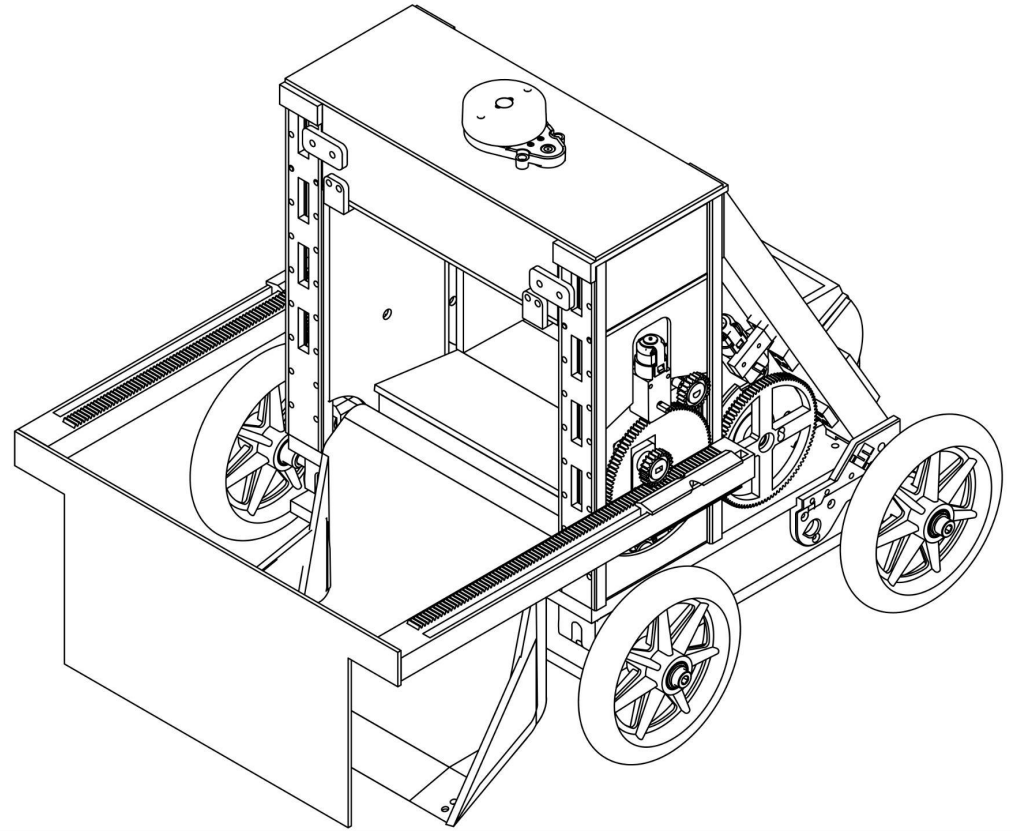


I/O & User Interface

Design Tradeoffs

- Cost vs Durability
 - Since we want to keep costs low by using 3D printing the durability of the robot is lower.
- Autonomy vs Safety
 - We wanted our robot to be safe around humans so it required more sensors in order to detect people and objects.

Prototype



Testing

Software Test Plans

- Manual Testing
- Functional Testing
- Usability Testing
- Integration Testing

Embedded System Test Plans

- Unit testing
- Integration testing
- System Testing

Machine Learning Testing

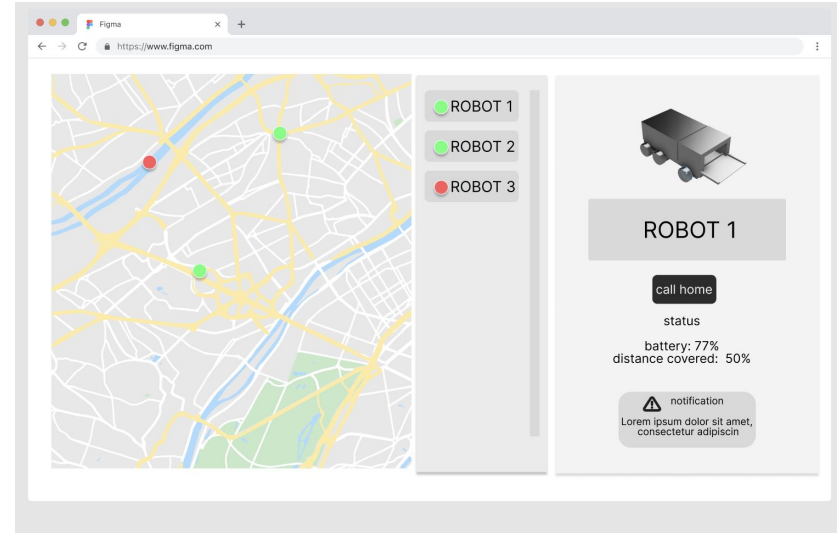
- Accuracy Test
- Inference Latency Benchmarks

Implementation

Prototype Evaluation

What we are looking for:

- Usability
- Average Time Spent Cleaning
- Battery Life
- Object Detection Accuracy



Concept of User Interface (Not our product's GPS)

Challenges

- Time Constraints
 - Two semesters of work on this project while taking other classes.
- Budget Constraints
 - We set our budget to be \$500 and wanted to use 3D printing for as much of the robot as possible.
- Working From Scratch
 - We wanted to build this project from scratch to get the full design experience from the hardware to the software.

Conclusions & Lessons Learned

- Prototype Early
 - An earlier prototype can provide useful information on how these systems can work in the real world.
- Avoid Scope Creep
 - Even though adding more to the project can seem like a good idea, it can also lead to a longer project.
- Test Systems Frequently
 - Testing after each change in design would ensure that the new change is exactly what we want.

Future

- Fault Tolerance
 - This would allow the robot to continue operating even if one part has a fault.
- Custom PCB design
 - Use of PCBs to make our connections instead of breadboards and wires would make our system more durable.
- Detection of moving objects
 - Adding more objects that the robot can detect would allow us for more advanced mapping and object avoidance.

Questions